

SKILLBRIDGE AI

AI-Powered Career Intelligence

Your Career, Smarter.

Hackathon Project Documentation

Alibaba Cloud AI Hackathon Pakistan 2026

Theme: **AI for Pakistan's Future**

Core Workflow

Profile / Resume → Skill Gap → Career Roadmap → Job Analysis → Market Insights → Interview Preparation

A practical AI platform that helps students and early-career professionals understand their skills, close market gaps, and prepare for real opportunities.

Prepared for Judging Panel · Confidential

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1 Project Overview

SkillBridge AI is an AI-powered career intelligence platform designed for students and early-career professionals. The platform helps users understand their existing skills, identify missing skills for their target career, analyze job requirements, explore market demand, build a personalized career roadmap, and prepare for interviews.

Instead of providing generic career advice, SkillBridge AI connects a user's profile with structured career and job requirements to deliver actionable, prioritized insights.

Core Workflow



Profile / Resume → Skill Gap → Career Roadmap → Job Analysis → Market Insights → Interview Preparation

2 Problem Statement

Students and fresh graduates in Pakistan (and globally) often face a structural gap between what they know and what the job market expects. A typical student may have:

- A resume but no clear understanding of their skill gaps.
- A target career but no structured learning path.
- The required technical skills but limited knowledge of job-specific requirements.
- Multiple skills to learn but no clear priority or sequencing.
- Difficulty assessing readiness for a particular role or job description.

Existing career platforms typically offer isolated services—job boards, resume builders, or course catalogs. The real challenge is connecting these pieces into one practical, end-to-end career intelligence workflow.

3 Our Solution

SkillBridge AI unifies profile intelligence, skill analysis, job requirements, market insights, and interview preparation into a single coherent platform. Users can:

1. Create or update their career profile.
2. Analyze their resume using AI (structured extraction).
3. Identify current skills and compare them against target-role requirements.
4. Discover skill gaps and generate learning priorities plus a career roadmap.
5. Analyze a career goal or paste a full job description for match scoring.
6. Calculate job-specific skill matching (required / preferred / matched / missing).
7. Generate relevant interview questions based on the target role.
8. Explore job-market insights derived from available demand signals.

Goal: Help users understand where they are, where the market needs them to be, and what they should do next—clearly and actionably.

4 Key Features

4.1 Resume Intelligence

Users upload a resume (PDF). The system extracts structured career information—name, education, experience, skills, and target-role signals where present—while avoiding unsupported assumptions. AI handles natural-language understanding; deterministic logic performs normalization and validation.

4.2 Skill Gap Analysis

SkillBridge AI compares the user's current skills against the requirements of their selected target role. The system surfaces:

- Current skills
- Required skills
- Missing skills
- Learning priorities

This converts a broad career goal into specific, measurable areas for improvement.

4.3 Career Roadmap

Identified skill gaps are transformed into a structured preparation roadmap. Rather than a vague instruction such as “You need to learn PyTorch,” the platform organizes what should be learned, in what order, and with what priority.

4.4 Job Description Analysis

Users can describe a target role or paste a complete job description. The system extracts requirements and compares them against the user's profile, producing:

- Required skills · Preferred skills
- Matched skills · Missing skills
- Match score · Learning priorities
- Preparation roadmap · Interview questions

4.5 Market Insights

The Market Insights module analyzes available job-market data to highlight frequently appearing skills, relevant roles, skill demand signals, user–market matches, and potential gaps—giving users a broader view of market expectations.

4.6 Interview Preparation

SkillBridge AI generates interview questions grounded in the user's target role and the requirements extracted from job analysis. This closes the loop from “I know what I need to learn” to “I can prepare to demonstrate it.”

5 AI Implementation

SkillBridge AI uses **Alibaba Cloud DashScope / Qwen** for intelligent analysis of unstructured career information. AI is applied primarily where natural-language understanding is required.

AI Responsibilities

- Resume information extraction
- Skill extraction from free-text
- Job requirement extraction
- Required / preferred skill identification
- Career-related text analysis

Critically, SkillBridge AI does **not** rely solely on generative AI for core decision logic. The application follows a hybrid intelligence architecture:

Hybrid Architecture

Unstructured Text → Qwen / DashScope → Structured AI Response
→ Validation → Deterministic Career Logic → Career Insights

Structured application logic handles skill normalization, matching, gap calculation, prioritization, and roadmap generation. This separation yields more predictable, consistent results while still leveraging AI's strength in understanding natural language.

6 System Architecture



Technology Stack

Layer	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python, FastAPI
Database	SQLite
ORM	SQLAlchemy
Validation	Pydantic
AI	Alibaba Cloud DashScope / Qwen
PDF Processing	PyPDF
Server	Uvicorn
Deployment	Render

7 Backend Architecture

The FastAPI backend provides the core application services. Major components include:

- Profile Management** — Create, retrieve, update, and delete user profiles.
- Resume Analysis** — Process uploaded resumes and extract structured career information.
- Skill Gap Engine** — Compare current skills against target-role requirements.
- Job Analysis Engine** — Extract job requirements and calculate user–job compatibility.
- Market Analysis** — Process job-market signals and identify relevant skills and roles.
- AI Service Layer** — Communicate with Qwen/DashScope and provide local fallback intelligence.

8 API Overview

Method	Endpoint	Purpose
GET	/	API status
POST	/profile	Create profile
GET	/profile/{id}	Retrieve profile
PUT	/profile/{id}	Update profile
DELETE	/profile/{id}	Delete profile

Method	Endpoint	Purpose
GET	/profile/{id}/skill-gap	Calculate skill gap
POST	/profile/{id}/job-analysis	Analyze career goal or JD
GET	/profile/{id}/market-analysis	Analyze market demand
POST	/resume/analyze	Analyze uploaded resume

9 Security & Validation

Security and input validation were designed into the application from the start.

Input Validation

- Profile fields (type, length, required constraints)
- Resume file type and size limits
- Job-analysis input sanitization
- Structured AI response validation before use
- API request data via Pydantic schemas

AI Input Security

Job descriptions and other externally supplied text are treated as untrusted input. The system includes protection against prompt-injection-style content and does not treat instructions embedded inside submitted job descriptions as system instructions.

Secrets Management

API credentials are stored exclusively through environment variables and are never hard-coded or committed to the repository.

Structured AI Responses

All AI-generated outputs are validated against expected schemas before being consumed by application logic, reducing the risk of malformed or adversarial responses propagating.

10 Testing

Automated testing was performed throughout development to verify core functionality and prevent regressions.

Current Test Result: 8 / 8 tests passed

Coverage includes core application functionality, job analysis, and the market analysis pipeline.

pytest

Manual Testing Coverage

- Profile creation & updating
- Resume analysis
- Skill-gap analysis & career roadmap
- Career-goal analysis & job-description analysis
- Market insights
- Interview preparation
- Validation and error handling

11 Demonstration Workflow

For the live demonstration, SkillBridge AI can be presented as one continuous user journey:

Step 1 — Profile

Create or load a student / fresh-graduate profile.

Step 2 — Skills

Show the user's current skills and declared target role.

Step 3 — Skill Gap

Identify required skills the user currently lacks.

Step 4 — Roadmap

Demonstrate conversion of gaps into learning priorities.

Step 5 — Job Analysis

Paste a sample job description or describe a target role.

Step 6 — Job Fit

Display Match Score, Required / Matched / Missing Skills, Learning Priorities.

Step 7 — Interview Prep

Generate interview questions based on the analyzed role.

This demonstrates the complete journey from self-assessment to job preparation.

12 Example Use Case

User Profile

A final-year Computer Science student with experience in:

- Python
- Machine Learning
- SQL
- Git

Target Role: AI Engineer

SkillBridge Analysis

The system identifies additional requirements such as Deep Learning, PyTorch, TensorFlow, LLMs, and RAG. It then converts these gaps into prioritized learning items and enables the user to compare their profile against a specific AI Engineer job description—providing a far more practical answer than “Learn AI.”

13 Impact

SkillBridge AI is designed to reduce the gap between **what a candidate knows** and **what the job market requires**. The platform guides users through a practical cycle:

Understand → Identify → Prioritize → Prepare → Apply

Potential Users

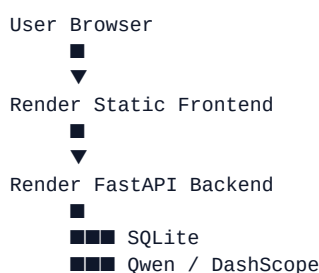
- University students
- Fresh graduates
- Early-career professionals
- Career switchers
- Self-taught developers

The platform does not guarantee employment. Its purpose is to provide structured information and actionable preparation guidance that users can act on immediately.

14 Deployment

SkillBridge AI is deployed using a two-part architecture on Render:

- **Frontend** — Static HTML/CSS/JavaScript hosted on Render.
- **Backend** — FastAPI application hosted as a Render Web Service.



The current SQLite implementation is appropriate for the hackathon prototype. A managed production database (e.g., PostgreSQL) can be introduced for long-term persistence and scaling.

15 Project Structure

```
SkillBridge-AI/
├── backend/
│   ├── main.py
│   └── models.py
```

```
■ ■■■ schemas.py
■ ■■■ ai_service.py
■ ■■■ skill_gap.py
■ ■■■ resume_parser.py
■ ■■■ ...
■■■ frontend/
■ ■■■ index.html
■ ■■■ app.js
■ ■■■ styles.css
■■■ tests/
■ ■■■ test_core.py
■ ■■■ test_job_analysis.py
■ ■■■ test_market_pipeline.py
■■■ docs/
■ ■■■ SkillBridge_AI_Documentation.pdf
■■■ README.md
■■■ requirements.txt
■■■ .gitignore
```

16 Future Scope

SkillBridge AI can be extended well beyond the current hackathon prototype. Potential improvements include:

- Real-time job-market integrations (live job feeds)
- Expanded Pakistani job-market intelligence and localization
- Multilingual career guidance (Urdu / regional languages)
- AI-powered mock interviews with feedback
- Personalized learning-resource recommendations
- Career progress tracking and milestone alerts
- More advanced candidate–job matching algorithms
- Managed production database (PostgreSQL)
- Additional AI model providers and ensemble strategies
- Deeper labor-market analytics and trend forecasting

17 Conclusion

SkillBridge AI transforms career planning from a fragmented process into a connected intelligence workflow. It combines:

Personal Profile + Skills + AI Understanding + Job Requirements + Market Insights + Career Planning

into one practical platform. The core vision is simple:

Your Career, Smarter.

SkillBridge AI helps students and early-career professionals understand their current position, identify the skills they need, and turn that information into a practical path toward their career goals.

18 Project Links

Resource	URL / Location
Live Demo	YOUR_RENDER_FRONTEND_URL
GitHub Repository	YOUR_GITHUB_REPOSITORY_URL
Documentation	YOUR_GITHUB_DOCUMENTATION_PDF_URL
Demo Video	YOUR_DEMO_VIDEO_URL

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